

Introduction

The scalar curvature of a Riemannian manifold (M, g) is a smooth function $\text{scal}_g: M \rightarrow \mathbb{R}$. Roughly speaking, the value $\text{scal}_g(x)$ at a point $x \in M$ measures how much the volume of a small geodesic ball around x deviates from that of a Euclidean ball of the same radius. While scalar curvature is the weakest of the standard curvature conditions for a Riemannian manifold, the presence of *positive* scalar curvature can still have strong topological implications. The archetypal example is the celebrated Gauß-Bonnet Theorem:

Theorem (Gauß-Bonnet). *Let (M, g) be a closed Riemannian 2-manifold. Then*

$$\int_M \text{scal}_g \, dA = 4\pi\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M .

So any closed surface carrying a metric of positive scalar curvature must be diffeomorphic to either the sphere S^2 or $\mathbb{R}P^2$. For non-compact complete manifolds, positive scalar curvature alone may be too weak a condition for such rigidity results though. The problem is that scal can still tend to zero quite rapidly at infinity. However, if the rate of decay at infinity can be controlled, positive scalar curvature still has profound consequences. For example, Chen has recently shown a decomposition result for 3-manifolds under the assumption that a metric of positive scalar curvature with controlled decay at infinity exists [Chen2025]. This naturally raises the question of how fast the scalar curvature of a given complete Riemannian manifold X decays at infinity. One result in this direction is the following theorem by Gromov [?, p. 15f.].

Theorem (Enlargeable Decay Theorem). *Let (X, g) be a complete Riemannian manifold and $V \rightarrow M$ a 2-dimensional real vector bundle over an enlargeable manifold M . Assume that*

- (i) *There exists a proper map $f: X \rightarrow V$, $\deg(f) \neq 0$,*
- (ii) *The total space of the circle bundle corresponding to V is enlargeable.*

Then

$$\min_{B_R(x_0)} \text{scal}_g \leq \frac{8\pi^2}{(R - R_0)^2}$$

for some $R_0 = R_0(X, x_0)$ and all $R > R_0$, where $B_R(x_0)$ is the ball of radius R around a point $x_0 \in M$.

Condition (ii) is always fulfilled if the bundle $V \rightarrow M$ is trivial (compare Lemma ?? ??), which led Gromov to pose the following question, here written using the notation introduced above.

Question ([?, p. 16]). “What happens to nontrivial bundles $V \rightarrow M$? [...] In general, the examples in [?] indicate a possibility of non-enlargeable circle bundles over enlargeable M ; yet, it seems hard(er) to find such examples, where the corresponding V would admit complete metrics with $\text{scal} \geq \sigma > 0$.”

Note that the existence of such a bundle $V \rightarrow M$ would provide counterexamples to the above decay theorem if condition (ii) is dropped entirely.

As anticipated by Gromov, examples of non-enlargeable circle bundles over enlargeable manifolds do in fact exist. Bundles with this behaviour have recently been constructed in a paper by Kumar and Sen [?] by utilizing tools from symplectic geometry. A very rough sketch of the construction can be given as follows. Take M to be a so-called Donaldson divisor of a suitable closed symplectic manifold, e.g., the torus T^{2n} , $n \geq 3$. Such a Donaldson divisor is a certain codimension 2 symplectic submanifold $M^{2n-2} \subseteq T^{2n}$, which in some cases can be shown to be enlargeable. Now define a circle bundle $E := \partial\nu(M)$ over M as the boundary of a tubular neighbourhood $\nu(M) \subseteq T^{2n}$ of M . The authors then proceed to show that E admits a metric of positive scalar curvature by the following argument. Consider the nullbordism $W := T^{2n} - \nu(M)$ obtained by removing the tubular neighbourhood from the ambient manifold. Now explicitly construct a Morse function $\psi: W \rightarrow \mathbb{R}$ with critical points of index at most n , which shows that E is obtained from the empty set by surgeries of codimension at least $n \geq 3$. Thus, E admits a metric of positive scalar curvature by the Surgery Theorem (see Theorem 1.7) and so in particular it is not enlargeable (see Theorem ??).

In this thesis, we take a different approach and provide a much more direct construction of such bundles drawing only on general topological concepts, without invoking symplectic methods. This approach, however, is limited to the totally non-spin case. We then apply our results to obtain examples of 2-dimensional vector bundles over enlargeable manifolds, where the total space admits a metric of uniformly positive scalar curvature, thereby providing an affirmative answer to Gromov's question. Note that

The thesis is organized as follows. Chapter 1 serves as a review of principal bundles, classifying spaces and characteristic classes and lays the foundation for the subsequent chapters. In Chapter 2, we briefly introduce the reader to the theory of positive scalar curvature and prove two existence theorems, which are later applied in the main construction. Chapter 3 provides a brief discussion of the concept of enlargeability. With the necessary background established, we then present the main results discussed above in Chapter 4. Finally, Chapter 5 discusses some further questions regarding the preceding construction.

1 Positive Scalar Curvature

The purpose of this chapter is twofold. The first section provides a brief introduction to positive scalar curvature, following [?] and [?]. In the second section, we prove two concrete existence results for metrics of positive scalar curvature, which will be used in the construction presented in Chapter 4.

1.1 A Brief Introduction

Let M^n be an n -dimensional manifold. Given any Riemannian metric g on M , we can associate to it a smooth function $\text{scal}_g: M \rightarrow \mathbb{R}$ called the *scalar curvature* of M . In the case where M is an embedded surface in $\mathbb{R}^3$, scal_g equals twice the classical Gaussian curvature $K: M \rightarrow \mathbb{R}$. So in this case, the scalar curvature has a clear geometric interpretation, illustrated below.

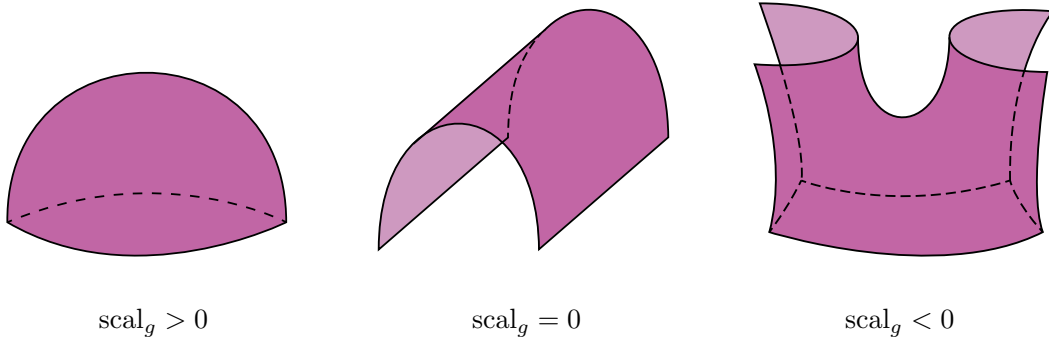


Figure 1.1: Scalar curvature in dimension 2.

In dimension 2, scalar curvature is the only intrinsic notion of curvature. However, in dimensions $n \geq 3$, there exists a whole hierarchy of different curvatures, the most prominent of which are sectional curvature, Ricci curvature and scalar curvature. The sectional curvature is the strongest notion and determines the entire geometry of M . In this thesis, we are concerned with the scalar curvature, which is the weakest among the three. In most textbooks on Riemannian geometry, it is defined by taking twice the trace of the Riemannian curvature tensor. While this definition is best suited for explicit computations, it provides no further intuition on what scalar curvature is. Since we have no need for doing such explicit computations, we give a more geometric definition, which essentially extends our intuition in the 2-dimensional case to arbitrary dimensions.

Definition 1.1. Let (M^n, g) be an n -dimensional Riemannian manifold. The scalar curvature $\text{scal}_g(x) \in \mathbb{R}$ at the point $x \in M$ is defined as the value appearing in the third coefficient of the Taylor expansion

$$\frac{\text{vol}(B_r(x))}{\text{vol}(B_r^{\text{eucl.}}(0))} = 1 - \frac{\text{scal}_g(x)}{6(n+2)}r^2 + \dots,$$

where $\text{vol}(B_r(x))$ denotes the volume of a geodesic ball in M of radius r centered at x , and $\text{vol}(B_r^{\text{eucl.}}(0))$ denotes the volume of a ball of the same radius r in standard Euclidean space $\mathbb{R}^n$.

So, intuitively, $\text{scal}_g(x) > 0$ means that the volume of a ball in M around x of sufficiently small radius r is smaller than that of a Euclidean ball of the same radius. Similarly, if $\text{scal}_g(x) < 0$, the volume of the ball in M is larger, and if $\text{scal}_g(x) = 0$, the volumes are (asymptotically) identical. Note that this is in accordance with Figure 1.1.

Example 1.2. (i) We immediately obtain from the definition that the scalar curvature of $\mathbb{R}^n$ with the usual Euclidean metric is zero.

(ii) If (M, g) is one-dimensional, the volume of a geodesic ball of radius r is simply $2r$. Therefore, $\text{scal}_g = 0$.

(iii) The scalar curvature of the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ can be computed to equal $n(n-1)$.

(iv) Let (M_1, g_1) and (M_2, g_2) be Riemannian manifolds. Then, for the scalar curvature of the product $(M_1 \times M_2, g_1 + g_2)$ equipped with the product metric $g_1 + g_2$, one obtains $\text{scal}_{g_1+g_2}(x_1, x_2) = \text{scal}_{g_1}(x_1) + \text{scal}_{g_2}(x_2)$ for $x_i \in M_i$. We refer the reader to any classical textbook on Riemannian geometry for a proof of these last two examples.

Broadly speaking, the study of scalar curvature falls into two categories: *rigidity* results and *flexibility* results. As the title of this chapter suggests, one is primarily concerned with results about *positive* scalar curvature. The reason for this is the following theorem. See, for example, [?].

Theorem 1.3. *Let M^n be a closed connected manifold of dimension $n \geq 3$, and let $f: M \rightarrow \mathbb{R}$ be a smooth function.*

(i) *If there exists a point $p \in M$ with $f(p) < 0$, then M admits a metric g with scalar curvature $\text{scal}_g = f$.*

(ii) *If M admits a metric whose scalar curvature is non-negative but not identically zero, then M admits another metric g with scalar curvature $\text{scal}_g = f$.*

Part (i) of the theorem asserts that the only scalar curvature condition from which we can expect any kind of obstruction results is non-negativity. Moreover, aside from the scalar-flat case (where the scalar curvature is identically zero), part (ii) shows that studying non-negative scalar curvature yields nothing new beyond what is already captured by positive scalar curvature. In the following, we refer to a metric g on M whose scalar curvature is everywhere positive as a **PSC-metric**.

The reason why the study of positive scalar curvature has attracted particular attention compared to other curvature conditions, such as positive Ricci or sectional curvature, is that it strikes an exceptionally good balance between the rigidity and flexibility aspects. While positive Ricci or sectional curvature has substantial topological implications, this also makes it significantly more difficult to construct metrics with these curvature conditions. For example, the Bonnet-Myers theorem states that a positive lower bound on the Ricci curvature of a complete Riemannian manifold M already implies that M must be compact with finite fundamental group. For PSC-metrics, however, there are existence results under quite general assumptions, which we will discuss below. First, let us consider some examples illustrating the rigidity aspects of positive scalar curvature. The most prominent obstruction to the existence of a PSC-metric comes from spin geometry. Let (M^n, g) be a closed Riemannian spin manifold with n even. Then the spin structure and the metric g determine a Hermitian vector bundle $S \rightarrow M$ together with a compatible connection ∇^S and a self-adjoint first-order differential operator $D: \Gamma(S) \rightarrow \Gamma(S)$ acting

on sections of S . Moreover, there exists a canonical splitting $S \cong S^+ \oplus S^-$ with respect to which D splits as

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}.$$

We call $S \rightarrow M$ the *spinor bundle* of M and D the *Dirac operator*. One of the central motivations behind constructing this operator was to find a first-order differential operator squaring to the Laplacian. The spinor bundle S carries additional structure making it possible to define such an operator. However, its square agrees with the Laplacian only up to an error term determined by the curvature of M . More precisely, the *Schrödinger–Lichnerowicz formula* holds:

$$D^2 = \Delta^S + \frac{1}{4} \text{scal}_g,$$

where $\Delta^S: \Gamma(S) \rightarrow \Gamma(S)$ is the connection Laplacian determined by ∇^S . Given any vector bundle $E \rightarrow N$ over a closed Riemannian manifold N carrying a fiber metric and a compatible connection, it is a general fact that the induced connection Laplacian is a non-negative operator with respect to the L^2 -inner product $\langle \cdot, \cdot \rangle$ on $\Gamma(E)$ induced by the fiber metric and the volume form of N . Applying this to S , let $\psi \in \Gamma(S)$. Then

$$0 \leq \langle D\psi, D\psi \rangle = \langle D^2\psi, \psi \rangle = \langle \Delta^S\psi, \psi \rangle + \frac{1}{4} \int_M \text{scal}_g \|\psi\|^2,$$

where we used self-adjointness of D in the first equation and the Schrödinger–Lichnerowicz formula in the second. If $\text{scal}_g > 0$ and $\psi \neq 0$, the right-hand side is strictly positive, since Δ^S is a non-negative operator. But the right-hand side equals $\langle D\psi, D\psi \rangle = \|D\psi\|^2$, so this forces $D\psi \neq 0$. Hence the kernel of D must be trivial. In particular, the integer

$$\text{ind}(D) := \dim(\ker D^+) - \dim(\ker D^-),$$

known as the *index* of D , must vanish (one can show that the kernels of $D^\pm$ are finite-dimensional, so this is well defined). The major insight is that this index can also be computed in terms of a purely topological invariant of M , namely the so-called $\hat{A}$ -class $\hat{A}(TM) \in H^{4*}(M; \mathbb{Q})$ of the tangent bundle $TM \rightarrow M$, which is a certain characteristic class. The connection between the index of the Dirac operator and the $\hat{A}$ -class of the tangent bundle is established by the famous *Atiyah–Singer index theorem*, which states:

$$\text{ind}(D) = \langle \hat{A}(TM), [M] \rangle,$$

where the right-hand side is the evaluation of the cohomology class $\hat{A}(TM)$ against the fundamental class $[M]$ of M (for the components of $\hat{A}(TM)$ in degree $\neq n$, this is 0, so in particular $\langle \hat{A}(TM), [M] \rangle = 0$ if n is not divisible by 4). The number $\hat{A}(M) := \langle \hat{A}(TM), [M] \rangle \in \mathbb{Q}$ is also referred to as the $\hat{A}$ -genus of M . In summary:

Theorem 1.4 (Lichnerowicz, 1963). *Let (M^{4k}, g) be a closed Riemannian spin manifold with positive scalar curvature. Then $\hat{A}(M) = 0$.*

We refer the reader to [?] for an in-depth discussion of the above results. A further, more accessible and more geometric example for a rigidity result concerning positive scalar curvature is the following classical result by Llarull [?].

Theorem 1.5 (Llarull, 1998). *Let M^n be a closed connected Riemannian spin manifold of dimension $n \geq 2$. Suppose $\text{scal}_g \geq n(n-1)$ and let $f: M \rightarrow S^n$ be a smooth 1-Lipschitz map with $\deg(f) \neq 0$. Then f is an isometry.*

As we noted in Example 1.2 (iii), the constant $n(n-1)$ is equal to the scalar curvature of the n -sphere S^n with the standard round metric g_0 . Llarull then applied this theorem in the special case $M = S^n$ and $f = \text{id}$ to show that the only metric g on S^n with $\text{scal}_g \geq n(n-1)$ and $g \geq g_0$ is g_0 itself. Said differently, one cannot increase the curvature of the sphere without contracting distances. We conclude our discussion of scalar curvature rigidity results with another strongly geometric statement, which is an example for a so-called *band width estimate* [?].

Theorem 1.6 (Gromov, 2018). *Let $M = T^{n-1} \times [-1, 1]$ and let g be a Riemannian metric on M with scalar curvature bounded from below by a positive constant $\sigma > 0$. Then the distance between the two boundary components $\partial_- M = T^{n-1} \times \{-1\}$ and $\partial_+ M = T^{n-1} \times \{1\}$ satisfies*

$$\text{dist}_g(\partial_- M, \partial_+ M) \leq 2\pi \sqrt{\frac{n-1}{n\sigma}}.$$

Intuitively, we think of M as a cylinder (a ‘band’) over T^{n-1} with a left boundary component $\partial_- M$ and a right boundary component $\partial_+ M$. Then this theorem essentially states that the boundaries of such a band, which is highly positively curved, must be close together. There are various generalisations of Theorem 1.6 to allow for bands over other manifolds [?], [?]. Gromov even conjectures that T^{n-1} can be replaced by any manifold of dimension $n-1 \geq 5$ not admitting a PSC-metric [?, Conjecture C].

Let us now turn to the flexibility aspect of positive scalar curvature. The primary method for constructing PSC-metrics is the following theorem, known as the Surgery Theorem, which has been shown independently by Schoen and Yau [SY79] as well as Gromov and Lawson [GL80]. We refer the reader to the Appendix for a brief introduction to surgery and cobordism theory.

Theorem 1.7 (Gromov–Lawson, Schoen–Yau). *Let M be a closed manifold which admits a PSC-metric. Then any manifold M' obtained from M by surgery of codimension at least 3 also admits a PSC-metric.*

Remark 1.8. The construction in [?] of the PSC-metric on M' only changes the given PSC-metric on M locally near the embedding along which the surgery is performed. Therefore the same construction also applies to show the following version of the surgery theorem. Let M be a manifold with boundary and g a PSC-metric on M . Suppose that M' is obtained from M by surgery of codimension at least 3 along an embedding into the interior of M . Then M' admits a PSC-metric g' which agrees with g near the boundary $\partial M' \cong \partial M$. See also [?, Theorem 1.6].

In order to apply this theorem we need a way to control the dimension of the surgeries performed or equivalently the indices of handles in a given cobordism. This question is completely independent of scalar curvature and has for example been studied quite thoroughly in the context of the h -cobordisms theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall

Theorem 1.9 (Wall, 1971). *Let W^{n+1} be a cobordism from M_0 to M_1 , such that the inclusion is r -connected for $r \leq \dim(W) - 4$. Then W contains only handles of index $< \dim(W) - r$.*

Combining the above theorem with the Surgery Theorem 1.7, we can make the following observation. If M_0 admits a PSC-metric and the inclusion $M_1 \rightarrow W$ is 2-connected, then M_1 also admits a PSC-metric. In fact, using the conclusion of Theorem 1.9 it is possible to obtain an even stronger result. We say that a metric g on a manifold M with boundary is of *product form near the boundary*, if there is some collar neighbourhood $\iota: \partial M \times [0, 1) \rightarrow M$, such that $\iota^*g = g|_{\partial M} + dt^2$, where $g|_{\partial M}$ is the induced metric on ∂M .

Theorem 1.10 (extending psc to corobordism, Georg, Gaja). *Let W^{n+1} be a cobordism from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W$ is 2-connected. Then any PSC-metric on M_0 can be extended to a PSC-metric on W , which is of product form near the boundary.*

In view of the above discussion, one strategy to construct a psc metric on a given closed manifold M_1 of dimension at least 5 is to first find a cobordism W from some psc manifold M_0 to M_1 and then modify this cobordism by surgeries to obtain another cobordism W' from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is.

1.2 Existence Results for PSC-Metrics

In this section, we prove the following two existence results for PSC-metrics, one in the spin case and one in the totally non-spin case.

Theorem 1.11. *Let W^{n+1} be a cobordism from M_0 to M_1 , $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 , and that M_0 admits a PSC-metric. Then M_1 also admits a PSC-metric. In fact, the PSC-metric on M_0 can be extended to a PSC-metric on W , which is of product form near the boundary.*

Theorem 1.12. *Let W^{n+1} be an oriented cobordism from M_0 to M_1 , $n \geq 5$. Suppose that M_1 is connected and totally non-spin, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 , and that M_0 admits a PSC-metric. Then M_1 also admits a PSC-metric. In fact, the PSC-metric on M_0 can be extended to a PSC-metric on W , which is of product form near the boundary.*

The similarity between the two statements is no coincidence. In fact, both are a consequence of a more general result; see [?, Theorem 1.5]. The proofs presented here are essentially adaptations of the proof in [?] to the special cases at hand. Let us first prove Theorem 1.12.

Proof of Theorem 1.12. Without loss of generality, assume W to be connected as well. Otherwise, consider only the connected component containing M_1 . The crux of the proof is to find generators of $\pi_2(W, M_1)$ lifting to elements in $\pi_2(W)$ that are represented by embeddings with trivial normal bundle along which we can then perform surgery (compare Proposition ??). Ideally, we would like $\pi_2(W, M_1)$ to be a finitely generated group for this purpose, which is, however, not the case in general. Nevertheless, viewed as a $\mathbb{Z}[\pi_1(M_1)]$ -module, $\pi_2(W, M_1)$ is always finitely generated. We have included a detailed proof of this fact in Proposition ???. Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_2(W, M_1)$ be such a set of generators. We will now construct lifts in $\pi_2(W)$ with the desired properties. Let

$$F: W \rightarrow B := BSO(n+1)$$

be a classifying map for the tangent bundle TW and set $f := F \circ i: M_1 \rightarrow B$, where $i: M_1 \rightarrow W$ is the inclusion. Then f is a classifying map for $TM_1 \oplus \underline{\mathbb{R}}$ with $\underline{\mathbb{R}}$ the trivial line bundle. We claim that f is surjective on π_2 . Since $\pi_2(B) \cong \pi_1(SO(n+1)) \cong \mathbb{Z}_2$, it suffices to show that f is non-zero on π_2 . By Proposition ??, there exists an embedding $\iota: S^2 \rightarrow M_1$ such that $w_2(\iota^*TM_1) \neq 0$. We compute

$$w_2((f \circ \iota)^* \gamma_{n+1}^{\text{SO}}) = w_2(\iota^*(TM_1 \oplus \underline{\mathbb{R}})) = w_2(\iota^*TM_1) \neq 0.$$

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Therefore, $f \circ \iota$ represents a non-trivial element in $\pi_2(B)$, which shows the claim. Now consider the following diagram, where the row is exact, using the fact that $\pi_1(M_1) \rightarrow \pi_1(W)$ is an isomorphism.

$$\begin{array}{ccccccc} \pi_2(M_1) & \xrightarrow{i_*} & \pi_2(W) & \longrightarrow & \pi_2(W, M_1) & \longrightarrow & 0 \\ & \searrow f_* & \downarrow F_* & & & & \\ & & \pi_2(B) & & & & \end{array}$$

Using surjectivity of f_* , a diagram chase shows that there exist lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of the generators $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_2(W, M_1)$, which lie in the kernel of F_* . Moreover, since $n \geq 5$, the classes α_i can be represented by embeddings

$$\phi_i: S^2 \rightarrow W.$$

We show that the normal bundle ν_i of ϕ_i is trivial. Since W is oriented, it follows from Proposition ?? that $w_1(\phi_i^*TW) = 0$. Furthermore, $F \circ \phi_i$ is null-homotopic by construction, such that

$$w_2(\phi_i^*TW) = w_2((F \circ \phi_i)^* \gamma_{n+1}^{\text{SO}}) = 0.$$

Using $w_k(TS^2) = 0$ (see Example ?? ??), we compute

$$w_k(\nu_i) = w_k(\nu_i \oplus TS^2) = w_k(\phi_i^*TW) = 0$$

for $k = 1, 2$. Thus, Lemma ?? shows that ν_i is indeed trivial. We can therefore perform surgery to obtain a new cobordism W' from M_0 to M_1 with

$$\pi_2(W', M_1) = 0$$

and $\pi_1(W', M_1) \cong \pi_1(W, M_1) = 0$, see Proposition ?? ?? and Remark ??. In other words, the inclusion $M_1 \rightarrow W'$ is 2-connected. So, by Theorem 1.10, there exists a PSC-metric g on W' , which is of product form near the boundary. To construct a PSC-metric on the original cobordism W , observe that, since W' was obtained from W by surgeries of dimension 2, W can be obtained from W' by surgeries of codimension 3. Therefore, the Surgery Theorem 1.7 applies to construct a PSC-metric on W . Since this construction only changes the given metric g locally near the surgery embeddings, the resulting metric on W is still of product form near the boundary. $\square$

Proof of Theorem 1.11. As before, $\pi_2(W, M_1)$ is finitely generated as a $\mathbb{Z}[\pi_1(M_1)]$ -module. Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ be such generators. One could argue as in the proof of Theorem 1.12, replacing $BSO(n+1)$ by $B\text{Spin}(n+1)$ and observing that the corresponding f is trivially surjective on π_2 , as $\pi_2(B\text{Spin}(n+1)) \cong \pi_1(\text{Spin}(n+1)) = 0$. However, this just amounts to the following more direct argument. Choose lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ and represent each α_i by an embedding $\phi_i: S^2 \rightarrow W$. Then

$$w_k(\phi_i^*TW) = \phi_i^*w_k(TW) = 0$$

for $k = 1, 2$, since W is spin. The remainder of the proof proceeds as in Theorem 1.12. $\square$

In the main construction in Chapter ?? we will invoke Theorem 1.12 to obtain a PSC-metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

Lemma 1.13. (i) *The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

- (ii) A manifold containing a totally non-spin submanifold of codimension 0 is totally non-spin.
- (iii) If M is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \text{int}(M)$ in the interior of M .

Proof. For (i) let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_1, \quad \pi_2: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_2$$

the projections and let $\iota: \tilde{M}_1 \rightarrow \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii) suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold M . Let $\iota: N \rightarrow M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Then the differential $d\tilde{\iota}: T\tilde{N} \rightarrow T\tilde{M}$ is an isomorphism on fibers, such that $\tilde{\iota}^* T\tilde{M} \cong T\tilde{N}$. Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii) let $M_0 := M - D^n$. By the Seifert-van-Kampen Theorem the inclusion $i: M_0 \rightarrow M$ induces an isomorphism on π_1 . Therefore the universal cover $\tilde{M}_0$ of M_0 is given by the restriction of the universal cover $\tilde{M}$ of M to M_0 . Hence, $T\tilde{M}_0 = i^* T\tilde{M}$. By a Mayer-Vietoris sequence argument i is injective on $H^2(-; \mathbb{Z}_2)$ and consequently we obtain

$$w_2(T\tilde{M}_0) = i^* w_2(T\tilde{M}) \neq 0.$$

□